# The Mathematical Association of Victoria

**STUDENT NAME**_____

## Specialist Mathematics
## Trial Examination 1
Question and Answer Book

VCE Examination

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- Reading time is **15 minutes**
- Writing time is **1 hour**

**Materials supplied**

- Question and Answer Book of 12 pages
- Formula Sheet

**Instructions**

Students are **not** permitted to bring any technology (calculators or software), or notes of any kind, into the examination room.

Students are **not** permitted to bring mobile phones and/or any unauthorised electronic devices into the examination room.

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10 questions (40 marks)..... 2- 12

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## Instructions

- Answer **all** questions in the spaces provided.
- Write your responses in English.
- Unless otherwise specified, an **exact** answer is required for each question.
- In questions where more than one mark is available, appropriate working **must** be shown.
- Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.
- Take the **acceleration due to gravity** to have a magnitude $g$ m s$^{-2}$, where $g = 9.8$

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**Question 1** (3 marks)

Consider the polynomial $p(z) = 3z^4 - z^3 + z^2 - z - 2$ where $z \in C$
Given that a root of $p(z)$ is $-i$, find all solutions to the equation $p(z) = 0$.

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**Question 2** (3 marks)

Consider the differential equation $\dfrac{x}{\sqrt{9-y^2}} = \dfrac{1}{y}\dfrac{dx}{dy}$

Find $y$ in terms of $x$, given $x = 1$ when $y = 2\sqrt{2}$.

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**Question 3** (4 marks)

Consider the curve $2x^2y + 3y^2 - 4x^2 = 1$.

Find $\frac{d^2y}{dx^2}$ at $y = 1$.

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**Question 4** (10 marks)

**a.** State the equations of all asymptotes in the graph of $f(x)=\dfrac{2x^3}{x^2-4x+4}$ 1 mark

**b.** State the axial intercepts of the oblique asymptote. 1 mark

**c.** Find the stationary points on the graph of $f(x)=\dfrac{2x^3}{x^2-4x+4}$. 2 marks

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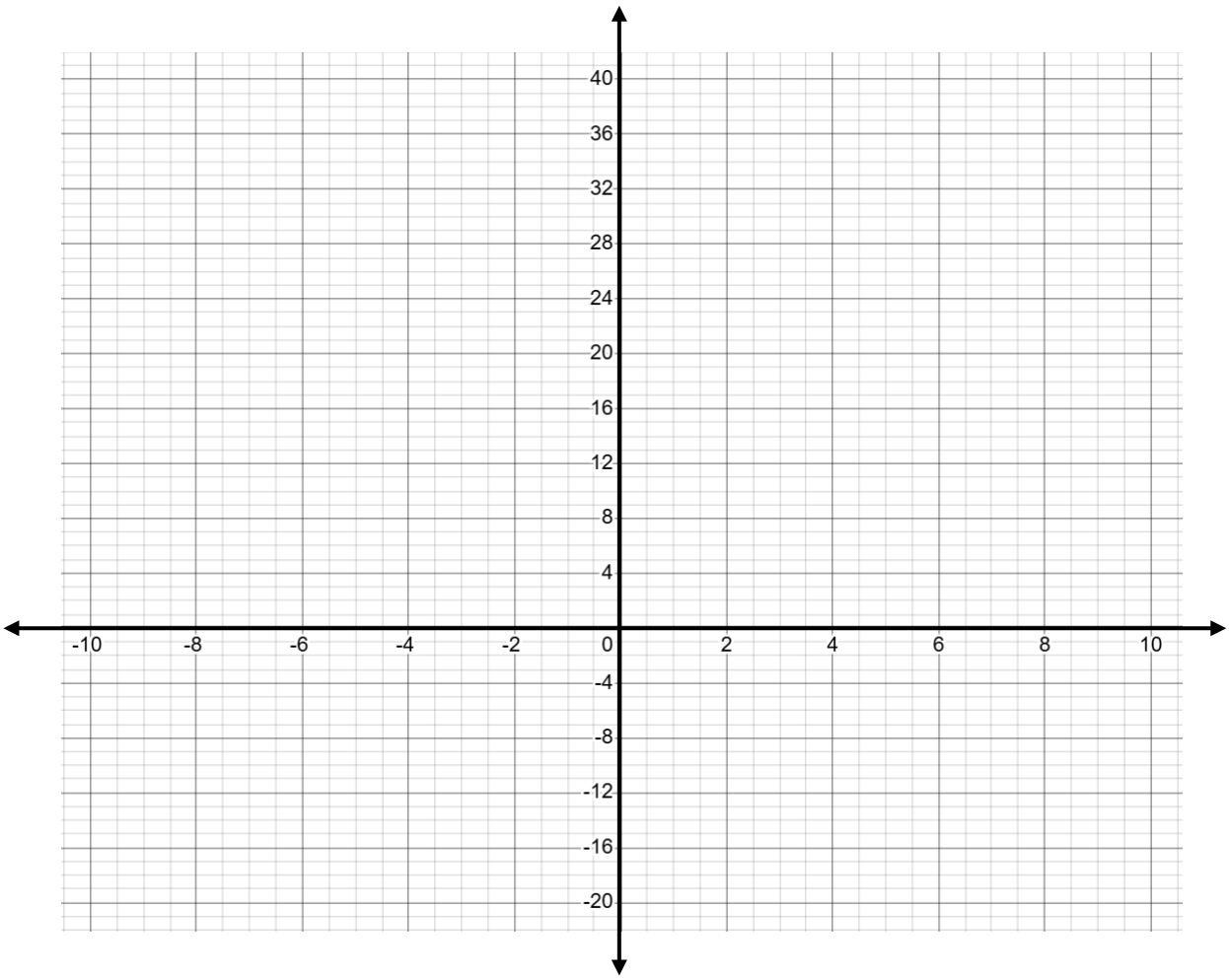
**d.** On the grid below sketch the graph of $y = f(x)$. Label asymptotes with their equations and coordinates of axial intercepts and stationary points. 3 marks

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**e.** Find the area under the graph of $f(x)$ between $x = 0$ and $x = 1$ 3 marks

**Question 5** (3 marks)

Use the method of *Proof by Contradiction* to show that $\sin(x) + \cos(x) \le \sqrt{2}$ for $x \in R$

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**Question 6** (3 marks)

Alex owns four identical departmental stores.

The mean and standard deviation of daily sales from each store are \$1000 and \$60, respectively. The mean and standard deviation of daily running costs of each store are \$400 and \$25, respectively. Assume that both the daily sales and running costs of a store are normally distributed.

Assuming that $b$ is the standard deviation, giving $b^2$ the variance, let $P \sim N\left(a, b^2\right)$ be the random variable that represents the net total daily profit, representing the total daily sales minus the total daily running costs, earned by Alex from all four stores. $a$ and $b$ are defined in dollars (\$).

**a.** Determine the values of $a$ and $b$. 2 marks

**b.** Use $\Pr(-1.5 < Z < 1.5) = 0.8664$ to determine the probability that Alex would earn a total profit more than $\$2595$ in a day from all four stores. Give your answer correct to four decimal places. 1 mark

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**Question 7** (3 marks)

Given that $x = \sin^{-1}\left(\frac{\sqrt{3}}{3}\right) + \cos^{-1}\left(\frac{\sqrt{5}}{5}\right)$, where $x \in \left(\frac{\pi}{2}, \frac{3\pi}{4}\right)$

Find $\cos(2x)$, expressing your answer as $\frac{a + b\sqrt{c}}{d}$, where $a, b, c, d \in Z$

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**Question 8** (4 marks)

*PQRS* is a rectangle with sides $\overrightarrow{PQ} = -5\underset{\sim}{i} + 10\underset{\sim}{j}$ and $\overrightarrow{QR} = 6\underset{\sim}{i} + 4\underset{\sim}{j}$.

*M* is a point on the side *RS* so that $\overrightarrow{SM} = -\frac{3}{2}\overrightarrow{RM}$

**a.** Find $\overrightarrow{PM}$. 2 marks

**b.** The diagonal *QS* intersects the line *PM* at point *N*.
If $\overrightarrow{PN} = m\overrightarrow{PQ} + (1-m)\overrightarrow{QR}$, find the value of $m$. 2 marks

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**Question 9** (4 marks)

The Cartesian equations of two planes $\Pi_1$ and $\Pi_2$ are as follows:

$$\Pi_1 : 2x + y - z = 10 \text{ and } \Pi_2 : x - 2y + z = 5$$

**a.** Given that vectors $\underset{\sim}{u} = a\underset{\sim}{i} - 2b\underset{\sim}{j} + 6\underset{\sim}{k}$ and $\underset{\sim}{v} = \underset{\sim}{i} + (a+1)\underset{\sim}{j} + (2-b)\underset{\sim}{k}$ both lie on the plane $\Pi_1$.
Determine the values of $a$ and $b$. 2 marks

**b.** Determine a vector equation of the line of intersection of the planes $\Pi_1$ and $\Pi_2$. 2 marks

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**Question 10** (3 marks)

Evaluate the definite integral $\int_0^{\frac{\pi}{2}} \left(2x^2 \cos^2(2x+\pi)\right)\, dx$

**End of examination questions**

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